

7 The grass *Oryza sativa* is grown as a food crop to produce rice. Rice plants are adapted to grow with their roots submerged in water.

(a) An experiment was carried out to investigate the development of aerenchyma tissue in the roots of rice plants.

- 10-day old seedlings were grown with their roots submerged in water.
- Some were grown in water that was kept oxygenated and others were grown in water that did not contain oxygen (deoxygenated).
- At 12 hour intervals, some seedlings from each group were removed and transverse sections of their roots were prepared and examined.
- The sections were cut at the same distances from the root tip to check for development.
- The percentage of root tissue that had developed into aerenchyma tissue was calculated.

The results are shown in Fig. 7.1.

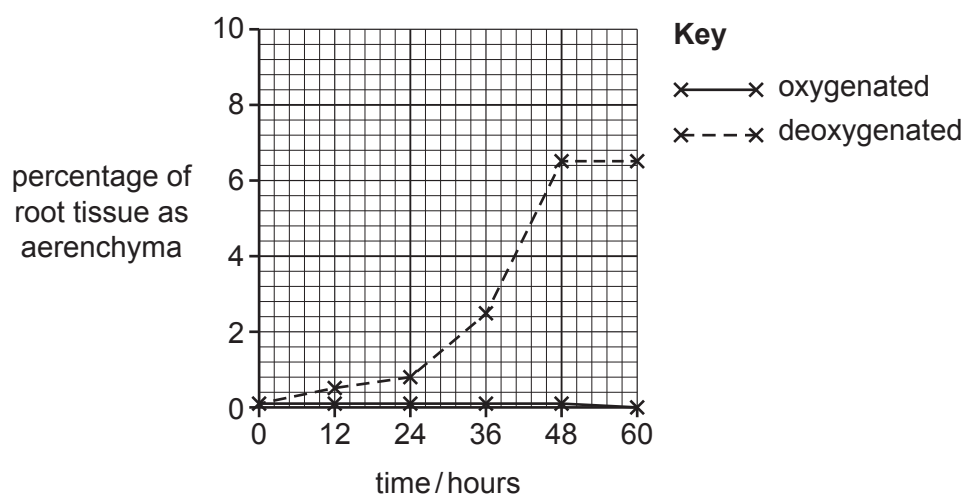


Fig. 7.1

(i) Describe the results shown in Fig. 7.1.

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- (ii) Describe the structure of aerenchyma **and** explain how this is an adaptation that allows roots of rice plants to be submerged in water.

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- (b) Describe **and** explain **two** other adaptations of rice plants to growing in flooded fields.

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[Total: 10]